

Quasar / Vue SCADA Components

1. GridMap.vue

```
<template>
  <q-page class="bg-dark text-white q-pa-md">
    <div class="row q-col-gutter-md">

      <div class="col-2">
        <q-card dark bordered class="bg-grey-10 full-height">
          <q-card-section>
            <div class="text-h6 text-primary">GRID MAP</div>
          </q-card-section>

          <q-list>
            <q-item clickable v-for="item in menu" :key="item">
              <q-item-section>{{ item }}</q-item-section>
            </q-item>
          </q-list>
        </q-card>
      </div>

      <div class="col-7">
        <q-card dark bordered class="bg-grey-10">
          <q-card-section class="row items-center justify-between">
            <div class="text-h6">Realtime Smart Grid</div>

            <div class="row q-gutter-sm">
              <q-chip color="green" text-color="white">Grid Stable</q-chip>
              <q-chip color="orange" text-color="white">82% Load</q-chip>
              <q-chip color="red" text-color="white">12 Active Alarms</q-
chip>
            </div>
          </q-card-section>

          <q-separator dark />

          <q-card-section>
            <div class="map-container">
              <div
                v-for="station in substations"
                :key="station.id"
                class="station"
                :style="{
                  left: station.x + '%',
                  top: station.y + '%',
                  background: station.color
                }">

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        >
      </div>
    </div>
  </q-card-section>
</q-card>
</div>

<div class="col-3">
  <q-card dark bordered class="bg-grey-10 q-mb-md">
    <q-card-section>
      <div class="text-h6">AI Forecasting</div>
    </q-card-section>

    <q-card-section>
      <q-linear-progress dark size="20px" :value="0.82" color="orange">
        <div class="absolute-full flex flex-center text-white">
          Load Forecast 82%
        </div>
      </q-linear-progress>

      <div class="q-mt-lg">
        <q-circular-progress
          show-value
          font-size="18px"
          :value="12"
          size="120px"
          color="red"
          track-color="grey-8"
        >
          12%
        </q-circular-progress>

        <div class="text-center q-mt-sm">
          Blackout Risk
        </div>
      </div>
    </q-card-section>
  </q-card>

  <q-card dark bordered class="bg-grey-10">
    <q-card-section>
      <div class="text-h6">AI Recommendations</div>
    </q-card-section>

    <q-list separator>
      <q-item>
        <q-item-section>
          Reduce load on TS-042
        </q-item-section>
      </q-item>
    </q-list>
  </q-card>

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        <q-item>
          <q-item-section>
            Inspect cooling system
          </q-item-section>
        </q-item>
      </q-list>
    </q-card>
  </div>
</div>
</q-page>
</template>

<script setup>
const menu = [
  'Dashboard',
  'Grid Map',
  'Transformers',
  'Analytics',
  'AI Forecasting',
  'Maintenance'
]

const substations = [
  { id: 'TS1', x: 20, y: 30, color: '#00ff88' },
  { id: 'TS2', x: 60, y: 40, color: '#ff8800' },
  { id: 'TS3', x: 45, y: 70, color: '#ff0033' }
]
</script>

<style scoped>
.map-container {
  position: relative;
  height: 700px;
  border-radius: 20px;
  background: radial-gradient(circle at center, #081421, #02060d);
  overflow: hidden;
}

.station {
  position: absolute;
  width: 18px;
  height: 18px;
  border-radius: 50%;
  box-shadow: 0 0 20px currentColor;
}
</style>

```

2. Transformers.vue

```
<template>
  <q-page class="bg-dark text-white q-pa-md">

    <div class="row q-col-gutter-md">

      <div class="col-8">
        <q-card dark bordered class="bg-grey-10">
          <q-card-section class="row justify-between items-center">
            <div class="text-h5">Transformers</div>

            <q-input dark dense outlined v-model="search" label="Search
Transformer" />
          </q-card-section>

          <q-table
            dark
            flat
            bordered
            :rows="rows"
            :columns="columns"
            row-key="id"
            class="bg-grey-10"
          >
            <template v-slot:body-cell-health="props">
              <q-td>
                <q-circular-progress
                  show-value
                  :value="props.row.health"
                  size="50px"
                  color="green"
                >
                  {{ props.row.health }}
                </q-circular-progress>
              </q-td>
            </template>
          </q-table>
        </q-card>
      </div>

      <div class="col-4">
        <q-card dark bordered class="bg-grey-10 q-mb-md">
          <q-card-section>
            <div class="text-h6">Transformer Health</div>
          </q-card-section>

          <q-card-section class="text-center">
            <q-circular-progress
              show-value
```

```

        font-size="20px"
        :value="78"
        size="180px"
        color="green"
      >
        78
      </q-circular-progress>
    </q-card-section>
  </q-card>

  <q-card dark bordered class="bg-grey-10">
    <q-card-section>
      <div class="text-h6">Maintenance Recommendations</div>
    </q-card-section>

    <q-list separator>
      <q-item>
        <q-item-section>
          Replace cooling fan – TR-0421
        </q-item-section>
      </q-item>

      <q-item>
        <q-item-section>
          Oil diagnostics required – TR-0783
        </q-item-section>
      </q-item>
    </q-list>
  </q-card>
</div>
</div>
</q-page>
</template>

<script setup>
import { ref } from 'vue'

const search = ref('')

const columns = [
  { name: 'id', label: 'ID', field: 'id' },
  { name: 'station', label: 'Substation', field: 'station' },
  { name: 'voltage', label: 'Voltage', field: 'voltage' },
  { name: 'load', label: 'Load', field: 'load' },
  { name: 'health', label: 'Health', field: 'health' }
]

const rows = [
  {
    id: 'TR-0421',
    station: 'Čakovec North',

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    voltage: '110/35kV',
    load: '82%',
    health: 72
  },
  {
    id: 'TR-0783',
    station: 'Varaždin West',
    voltage: '110/20kV',
    load: '64%',
    health: 91
  }
]
</script>

```

3. Analytics.vue

```

<template>
  <q-page class="bg-dark text-white q-pa-md">

    <div class="row q-col-gutter-md">

      <div class="col-3" v-for="card in kpis" :key="card.title">
        <q-card dark bordered class="bg-grey-10">
          <q-card-section>
            <div class="text-subtitle2 text-grey-5">{{ card.title }}</div>
            <div class="text-h4 text-primary q-mt-sm">{{ card.value }}</div>
          </q-card-section>
        </q-card>
      </div>
    </div>

    <div class="row q-col-gutter-md q-mt-md">

      <div class="col-8">
        <q-card dark bordered class="bg-grey-10">
          <q-card-section>
            <div class="text-h6">Realtime Grid Analytics</div>
          </q-card-section>

          <q-card-section>
            <div class="analytics-chart"></div>
          </q-card-section>
        </q-card>
      </div>

      <div class="col-4">
        <q-card dark bordered class="bg-grey-10 q-mb-md">
          <q-card-section>

```

```

        <div class="text-h6">Blackout Prediction</div>
    </q-card-section>

    <q-card-section class="text-center">
        <q-circular-progress
            show-value
            :value="12"
            size="150px"
            color="red"
        >
            12%
        </q-circular-progress>
    </q-card-section>
</q-card>

<q-card dark bordered class="bg-grey-10">
    <q-card-section>
        <div class="text-h6">Top Critical Assets</div>
    </q-card-section>

    <q-list separator>
        <q-item v-for="asset in assets" :key="asset.id">
            <q-item-section>
                {{ asset.id }}
            </q-item-section>

            <q-item-section side>
                <q-badge color="red">
                    {{ asset.risk }}%
                </q-badge>
            </q-item-section>
        </q-item>
    </q-list>
</q-card>
</div>
</div>
</q-page>
</template>

<script setup>
const kpis = [
    { title: 'Grid Load', value: '4.82 GW' },
    { title: 'Voltage Stability', value: '98.5%' },
    { title: 'Power Quality', value: '3.2%' },
    { title: 'Active Alarms', value: '12' }
]

const assets = [
    { id: 'TS-042', risk: 82 },
    { id: 'TR-0783', risk: 74 },
    { id: 'TS-091', risk: 69 }
]

```

```
]
</script>

<style scoped>
.analytics-chart {
  height: 500px;
  border-radius: 20px;
  background: linear-gradient(180deg, #081421, #02060d);
}
</style>
```

Router Example

```
const routes = [
  {
    path: '/grid-map',
    component: () => import('pages/GridMap.vue')
  },
  {
    path: '/transformers',
    component: () => import('pages/Transformers.vue')
  },
  {
    path: '/analytics',
    component: () => import('pages/Analytics.vue')
  }
]
```

Required Packages

```
npm install quasar
npm install echarts vue-echarts
npm install leaflet
npm install socket.io-client
```

Recommended Stack

Layer	Technology
Frontend	Vue.js

Layer	Technology
UI	Quasar
Maps	Leaflet
Charts	ECharts
Backend	FastAPI
MQTT	EMQX
Database	CrateDB
AI	PyTorch